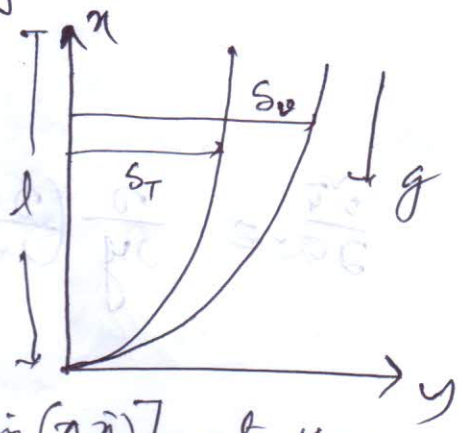
# (1) Boundary layer flow lecture (3)

## Mathematical Modelling 53 (2011) 1310–1319




$$\frac{\partial \bar{u}}{\partial \bar{x}} + \frac{\partial \bar{v}}{\partial \bar{y}} = 0$$

$$\rho\left(\bar{u}\frac{\partial \bar{u}}{\partial \bar{x}} + \bar{v}\frac{\partial \bar{u}}{\partial \bar{y}}\right) = \mu\frac{\partial^2 \bar{u}}{\partial \bar{y}^2} + \rho g\beta(T - T_\infty)$$

$$\bar{u}\frac{\partial T}{\partial \bar{x}} + \bar{v}\frac{\partial T}{\partial \bar{y}} = \frac{K}{\rho C_p}\frac{\partial^2 T}{\partial \bar{y}^2}$$

$\bar{u} = \bar{v} = 0$, $T = T_\infty + (T_w - T_\infty)\left[1 + A\sin\left(\pi\frac{\bar{x}}{l}\right)\right]$ at $y = 0$

$\bar{u} \to 0$, $T \to T_\infty$ as $\bar{y} \to \infty$.

### Non-dimensional variables

$$x = \frac{\bar{x}}{l}, \quad y = Gr^{1/4}\left(\frac{\bar{y}}{l}\right), \quad u = \frac{l}{\nu}Gr^{-1/2}\bar{u}, \quad v = \frac{l}{\nu}Gr^{-1/4}\bar{v}$$

$$\theta = \frac{T - T_\infty}{T_w - T_\infty}, \quad Gr = \frac{g\beta(T_w - T_\infty)l^3}{\nu^2}, \quad Pr = \frac{\nu}{\alpha}, \quad \nu = \frac{\mu}{\rho},$$

$$\alpha = \frac{K}{\rho C_p}$$

$$\frac{\partial \bar{u}}{\partial \bar{x}} = \frac{\partial}{\partial x}\left(\frac{\nu}{l}Gr^{1/2}u\right)\cdot\frac{\partial x}{\partial \bar{x}} = \frac{\nu}{l}Gr^{1/2}\frac{\partial u}{\partial x}\cdot\frac{1}{l} = \frac{\nu}{l^2}Gr^{1/2}\frac{\partial u}{\partial x}$$

$$\frac{\partial \bar{v}}{\partial \bar{y}} = \frac{\partial}{\partial y}\left(\frac{\nu}{l}Gr^{1/4}v\right)\cdot\frac{\partial y}{\partial \bar{y}} = \frac{\nu}{l}Gr^{1/4}\frac{\partial v}{\partial y}\cdot\frac{Gr^{1/4}}{l} = \frac{\nu}{l^2}Gr^{1/2}\frac{\partial v}{\partial y}$$

Cont$^n$. eq$^n$: $\frac{\partial \bar{u}}{\partial \bar{x}} + \frac{\partial \bar{v}}{\partial \bar{y}} = 0$

$$\Rightarrow \frac{\nu}{l^2}Gr^{1/2}\frac{\partial u}{\partial x} + \frac{\nu}{l^2}Gr^{1/2}\frac{\partial v}{\partial y} = 0$$

$$\Rightarrow \boxed{\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0}$$

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y}\left(\frac{\nu}{l} Gr^{1/2} u\right) \cdot \frac{\partial y}{\partial y} = \frac{\nu}{l} Gr^{3/2} \frac{\partial u}{\partial y} \cdot \frac{Gr^{1/4}}{l}$$

$$= \frac{\nu}{l^2} Gr^{3/4} \frac{\partial u}{\partial y}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right) \cdot \frac{\partial y}{\partial y} = \frac{\nu}{l^2} Gr^{3/4} \frac{\partial^2 u}{\partial y^2} \cdot \frac{Gr^{1/4}}{l}$$

$$= \frac{\nu}{l^3} Gr \frac{\partial^2 u}{\partial y^2}$$

$$\theta = \frac{T - T_\infty}{T_w - T_\infty} = \frac{T - T_\infty}{\Delta T}$$

momentum equation

$$\left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y}\right) = \nu \frac{\partial^2 u}{\partial y^2} + g\beta \Delta T \theta$$

$$\Rightarrow \frac{\nu Gr^{1/2}}{l} u \cdot \frac{\nu}{l^2} Gr^{1/2} \frac{\partial u}{\partial x} + \frac{\nu Gr^{1/4}}{l} v \cdot \frac{\nu}{l^2} Gr^{3/4} \frac{\partial u}{\partial y} = \nu \cdot \frac{\nu}{l^3} Gr \frac{\partial^2 u}{\partial y^2} + \theta \Delta T g\beta$$

$$\Rightarrow u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \frac{\nu^2}{l^3} Gr \cdot \frac{l^3}{Gr \cdot \nu^2} \frac{\partial^2 u}{\partial y^2} + \frac{\theta g\beta \Delta T l^3}{Gr\, \nu^2}$$

$$= \frac{\partial^2 u}{\partial y^2} + \theta \cdot \frac{\left(\frac{g\beta \Delta T l^3}{\nu^2}\right)}{Gr}$$

$$\Rightarrow \boxed{u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \frac{\partial^2 u}{\partial y^2} + \theta} \quad \text{——— (1)}$$

$$\frac{\partial T}{\partial x} = \frac{\partial}{\partial x}(\Delta T \theta) \frac{\partial x}{\partial x} = \frac{\Delta T}{l} \frac{\partial \theta}{\partial x}$$

$$\frac{\partial T}{\partial y} = \frac{\partial}{\partial y}(\Delta T \theta) \frac{\partial y}{\partial y} = \Delta T \frac{\partial \theta}{\partial y} \cdot \frac{Gr^{1/4}}{l} = \frac{\Delta T}{l} Gr^{1/4} \frac{\partial \theta}{\partial y}$$

$$\frac{\partial^2 T}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial T}{\partial y}\right) \frac{\partial y}{\partial y} = \frac{\partial}{\partial y}\left(\frac{\Delta T}{l} Gr^{1/4} \frac{\partial \theta}{\partial y}\right) \cdot \frac{Gr^{1/4}}{l}$$

$$= \frac{\Delta T}{l^2} Gr^{1/2} \frac{\partial^2 \theta}{\partial y}$$

Energy equation:

$$u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y} = \left(\frac{K}{\rho c_p} = \alpha\right)\frac{\partial^2 T}{\partial y^2}$$

$$\Rightarrow \frac{\nu Gr^{1/2}}{l} u \cdot \frac{\Delta T}{l}\frac{\partial \theta}{\partial x} + \frac{\nu Gr^{1/4}}{l} v \cdot \frac{\Delta T}{l} Gr^{1/4}\frac{\partial \theta}{\partial y} = \alpha \frac{\Delta T}{l^2} Gr^{1/2}\frac{\partial^2 \theta}{\partial y^2}$$

$$\Rightarrow u\frac{\partial \theta}{\partial x} + v\frac{\partial \theta}{\partial y} = \frac{\alpha \cancel{\Delta T} \cancel{Gr^{1/2}}}{\cancel{l^2}} \cdot \frac{\cancel{l^2}}{\nu \cancel{Gr^{1/2}} \cancel{\Delta T}}\frac{\partial^2 \theta}{\partial y^2}$$

$$= \frac{\alpha}{\nu}\frac{\partial^2 \theta}{\partial y^2}$$

$$\boxed{u\frac{\partial \theta}{\partial x} + v\frac{\partial \theta}{\partial y} = \frac{1}{Pr}\frac{\partial^2 \theta}{\partial y^2}} \quad (2)$$

Now we will introduce the parabolic transformation:

$$\psi = x^{3/4} f(x,\eta), \quad \eta = x^{-1/4} y, \quad \theta = \theta(x,\eta)$$

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}$$

$$u = \frac{\partial \psi}{\partial y} = x^{3/4}\frac{\partial f}{\partial \eta}\cdot\frac{\partial \eta}{\partial y} = x^{3/4} f' \cdot x^{-1/4} = x^{1/2} f'$$

$$\frac{\partial u}{\partial y} = x^{1/2}\frac{\partial f'}{\partial \eta}\frac{\partial \eta}{\partial y} = x^{1/2} f'' \cdot x^{-1/4} = x^{1/4} f''$$

$$\frac{\partial^2 u}{\partial y^2} = x^{1/4} f''' \cdot x^{-1/4} = f'''$$

$$v = -\frac{\partial \psi}{\partial x} = -\left[x^{3/4}\left\{\frac{\partial f}{\partial x}\frac{\partial x}{\partial x} + \frac{\partial f}{\partial \eta}\frac{\partial \eta}{\partial x}\right\} + f \cdot \frac{3}{4} x^{-1/4}\right]$$

$$= -x^{3/4}\frac{\partial f}{\partial x} + \frac{1}{4} x^{-1/4} y f' - \frac{3}{4} x^{-1/4} f$$

$$= -x^{3/4}\frac{\partial f}{\partial x} + \frac{1}{4x} x^{-1/4} y f' - \frac{3}{4} x^{-1/4} f$$

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}\left[x^{1/2} f'\right] = x^{1/2}\left\{\frac{\partial f'}{\partial x}\cdot\frac{\partial \eta}{\partial x} + \frac{\partial f'}{\partial \eta}\cdot\frac{\partial \eta}{\partial x}\right\} + f'\cdot\frac{1}{2}x^{-1/2}$$

$$= x^{1/2}\frac{\partial f}{\partial x} - \frac{x^{1/2}}{4x}\eta f'' + \frac{1}{2}x^{-1/2} f'$$

$$\Rightarrow \boxed{v = -x^{3/4}\frac{\partial f}{\partial x} + \frac{x^{3/4}}{4x}\eta f' - \frac{3}{4}x^{-1/4} f}$$

(1) $\Rightarrow$

$$x^{1/2} f'\left(x^{1/2}\frac{\partial f'}{\partial x} - \frac{x^{1/2}}{4x}\eta f'' + \frac{1}{2}x^{-1/2} f'\right)$$

$$+ \left(-x^{3/4}\frac{\partial f}{\partial x} + \frac{x^{3/4}}{4x}\eta f' - \frac{3}{4}x^{-1/4} f\right) x^{1/4} f'' = x f''' + \theta$$

$$\Rightarrow x f'\frac{\partial f'}{\partial x} - \cancel{\frac{1}{4}\eta f' f''} + \frac{1}{2} f'^2 - x f''\frac{\partial f}{\partial x} + \cancel{\frac{1}{4}\eta f' f''}$$

$$- \frac{3}{4} f f'' = f''' + \theta$$

$$\Rightarrow \boxed{f''' + \frac{3}{4} f f'' - \frac{1}{2} f'^2 + \theta = x\left(f'\frac{\partial f'}{\partial x} - f''\frac{\partial f}{\partial x}\right)}$$

Eng Energy

$$u\frac{\partial \theta}{\partial x} + v\frac{\partial \theta}{\partial y} = \frac{1}{Pr}\frac{\partial^2 \theta}{\partial y^2}$$

$$\boxed{\begin{aligned} \frac{\partial \theta}{\partial y} &= \frac{\partial \theta}{\partial x}\frac{\partial x}{\partial \eta} + \frac{\partial \theta}{\partial \eta}\cdot\frac{\partial \eta}{\partial y} \\ &= 0 + x^{-1/4}\theta' \\ \frac{\partial^2 \theta}{\partial y^2} &= x^{-1/2}\theta'' \end{aligned}}$$

$$\Rightarrow x^{1/2} f'\left(\frac{\partial \theta}{\partial x}\cdot\frac{\partial x}{\partial x} + \frac{\partial \theta}{\partial \eta}\frac{\partial \eta}{\partial x}\right) + \left(-x^{3/4}\frac{\partial f}{\partial x} + \frac{x^{3/4}}{4x}\eta f' - \frac{3}{4}x^{-1/4} f\right)$$

$$x^{-1/4}\theta' = \frac{1}{Pr}x^{-1/2}\theta''$$

$$\Rightarrow x^{1/2} f\frac{\partial \theta}{\partial x} + \cancel{x^{1/2} f'\frac{\partial \eta}{\partial x}\theta'} - x^{1/2}\frac{\partial f}{\partial x}\theta' + \cancel{\frac{1}{4x^{1/2}}\eta f'\theta'}$$

$$- \frac{3}{4}x^{-1/2} f\theta' = \frac{1}{Pr}x^{-1/2}\theta''$$

$$\Rightarrow x f\frac{\partial \theta}{\partial x} - x\frac{\partial f}{\partial x}\theta' - \frac{3}{4} f\theta' = \frac{1}{Pr}\theta''$$

$$\Rightarrow \frac{1}{Pr}\theta'' + \frac{3}{4} f\theta' = x\left(f'\frac{\partial\theta}{\partial\eta} - \theta'\frac{\partial f}{\partial\eta}\right)$$

<u>B.C</u>

$$f(x,0) = f'(x,0) = 0, \quad \theta(x,0) = 1 + \alpha \sin(\pi x)$$

$$f(x,0) \to 0, \quad \theta \to 0 \quad \text{as } \eta \to \infty.$$

## Numerical discretisation:

Let $f = v$, $f' = u$, $f'' = \frac{\partial u}{\partial y}$, $\theta' = \frac{\partial\theta}{\partial y}$, $\theta'' = \frac{\partial^2\theta}{\partial y^2}$

$$f''' + \frac{3}{4} f f'' - \frac{1}{2} f'^2 + \theta = x\left(f'\frac{\partial f'}{\partial\eta} - f''\frac{\partial f}{\partial x}\right)$$

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} + \frac{3}{4} v\frac{\partial u}{\partial y} - \frac{1}{2} u^2 + \theta = x\left(u\frac{\partial u}{\partial\eta} - \frac{\partial u}{\partial y}\frac{\partial v}{\partial\eta}\right)$$

$$\Rightarrow \frac{u_{ij+1} - 2u_{ij} + u_{ij-1}}{\Delta y^2} + \frac{3}{4} v_{ij}\frac{u_{ij+1} - u_{ij-1}}{2\Delta y} - \frac{1}{2} u_{ij} u_{ij} + \theta_{ij}$$

$$= x_i\left(u_{ij}\frac{u_{ij} - u_{i-1,j}}{\Delta x} - \frac{u_{ij+1} - u_{ij-1}}{2\Delta y}\frac{\partial v}{\partial x}\right)$$

$$\Rightarrow u_{ij+1} - 2u_{ij} + u_{ij-1} + \frac{3}{4} v_{ij}(u_{ij+1} - u_{ij-1})\left(\frac{\Delta y}{2}\right) - \frac{1}{2} u_{ij} u_{ij}\Delta y^2$$

$$+ \theta_{ij}\Delta y^2 = x_i\left(\frac{\Delta y^2}{\Delta x}\right) u_{ij}(u_{ij} - u_{i-1,j}) - x_i(u_{ij+1} - u_{ij-1})\frac{\partial v}{\partial\eta}\left(\frac{\Delta y}{2}\right)$$

$$\Rightarrow \left\{1 - \frac{3}{4} v_{ij}\left(\frac{\Delta y}{2}\right) - x\frac{\partial v}{\partial\eta}\left(\frac{\Delta y}{2}\right)\right\} u_{ij-1} + \left\{2 - \frac{1}{2} u_{ij}\Delta y^2 - x_i\frac{\Delta y^2}{\Delta x} u_{ij}\right\}$$

$$u_{ij} + \left\{1 + \frac{3}{4} v_{ij}\left(\frac{\Delta y}{2}\right) + x\frac{\partial v}{\partial\eta}\left(\frac{\Delta y}{2}\right)\right\} u_{ij+1} = -x\frac{(\Delta y)^2}{\Delta x} u_{ij} u_{i-1,j} - \theta_{ij}\Delta y^2.$$

$$\boxed{A_{ij}\, u_{ij-1} + B_{ij}\, u_{ij} + C_{ij}\, u_{ij+1} = D_{ij}} \quad (*)$$

where the matrix coefficients are

$$A_{ij} = 1 - \frac{3}{4} v_{ij}\left(\frac{\Delta y}{2}\right) - x_i \frac{\partial v}{\partial x}\left(\frac{\Delta y}{2}\right)$$

$$B_{ij} = -2 - \frac{1}{2} u_{ij} \Delta y^2 - x_i \frac{\Delta y^2}{\Delta x} u_{ij}$$

$$C_{ij} = 1 + \frac{3}{4} v_{ij}\left(\frac{\Delta y}{2}\right) + x \frac{\partial v}{\partial x}\left(\frac{\Delta y}{2}\right)$$

$$D_{ij} = -x_i \frac{(\Delta y)^2}{\Delta x} u_{ij}\, u_{i-1,j} - \theta_{ij} \Delta y^2$$

$$\frac{\partial v}{\partial x} = \frac{v_{ij} - v_{i-1,j}}{\Delta x}$$

We can solve (*) eq using Gaussian elimination method for $u_{ij}$.

~~Energ~~ Energy Eq$^n$

$$\frac{1}{Pr}\theta'' + \frac{3}{4} f\theta' = x\left(f' \frac{\partial \theta}{\partial x} - \theta' \frac{\partial f}{\partial x}\right)$$

$$\Rightarrow \frac{1}{Pr}\left(\frac{\theta_{ij+1} - 2\theta_{ij} + \theta_{ij-1}}{\Delta y^2}\right) + \frac{3}{4} v_{ij}\left(\frac{\theta_{ij+1} - \theta_{ij-1}}{2\Delta y}\right) = x\left\{u_{ij}\left(\frac{\theta_{ij} - \theta_{i-1,j}}{\Delta x}\right) - \left(\frac{\theta_{ij+1} - \theta_{ij-1}}{2\Delta y}\right)\frac{\partial v}{\partial x}\right\}$$

$$\Rightarrow \frac{1}{Pr}\left(\theta_{ij+1} - 2\theta_{ij} + \theta_{ij-1}\right) + \frac{3}{4} v_{ij}\left(\theta_{ij+1} - \theta_{ij-1}\right)\left(\frac{\Delta y}{2}\right)$$

$$= x\, u_{ij}\left(\theta_{ij} - \theta_{i-1,j}\right)\frac{\Delta y^2}{\Delta x} - \left(\theta_{ij+1} - \theta_{ij-1}\right)\frac{\partial v}{\partial x}\left(\frac{\Delta y}{2}\right) x$$

$$\Rightarrow \overbrace{\left\{\frac{1}{Pr} - \frac{3}{4} v_{ij} \frac{\Delta y}{2} - x \frac{\partial v}{\partial x}\left(\frac{\Delta y}{2}\right)\right\}}^{A_{ij}} \theta_{ij-1} + \overbrace{\left\{\frac{2}{Pr} - x \frac{\Delta y^2}{\Delta x} u_{ij}\right\}}^{B_{ij}} \theta_{ij}$$

$$+ \overbrace{\left\{\frac{1}{Pr} + \frac{3}{4} v_{ij} \frac{\Delta y}{2} + x \frac{\partial v}{\partial x}\left(\frac{\Delta y}{2}\right)\right\}}^{C_{ij}} \theta_{ij+1} = \underbrace{- x u_{ij} \theta_{i-1,j}}_{D_{ij}}$$

$$\Rightarrow \boxed{A_{ij}\, \theta_{ij-1} + B_{ij}\, \theta_{ij} + C_{ij}\, \theta_{ij+1} = D_{ij}}$$

$v$ is calculated from, $\boxed{f' = u} \Rightarrow v' = u$

Since $\boxed{v = f}$

$$\Rightarrow \frac{u_{ij} - u_{ij-1}}{\Delta y} = u_{ij}$$

$$\Rightarrow \boxed{u_{ij} = u_{ij-1} + u_{ij} \Delta y}$$

<u>Boundary Conditions:</u>

$u_{ij} = 0, \quad v_{ij} = 0 \qquad \theta_{ij} = 1 + \alpha \sin(\pi x)$ at $y = 0$ $(j = 1)$

$u = v = 0, \quad \theta = 0$ at $y \to \infty$ $\quad j = N$ (Maximum value of $j$)

$\psi = x f(x,y)$ $y = y$, $\theta = \theta(x,y)$

$$u = \frac{\partial \psi}{\partial y} = x\left(\frac{\partial f}{\partial x}\frac{\partial x}{\partial y} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial y}\right)$$

$$= x\frac{\partial f}{\partial y} = xf'$$

$$v = -\frac{\partial \psi}{\partial x} = -\frac{\partial}{\partial x}(xf) = -x\left(\frac{\partial f}{\partial x}\frac{\partial x}{\partial x} + \frac{\partial f}{\partial y}\cancel{\frac{\partial y}{\partial x}}^{0}\right) - f$$

$$= -x\frac{\partial f}{\partial x} - f$$

$$\frac{\partial u}{\partial y} = xf'', \quad \frac{\partial^2 u}{\partial y^2} = xf'''$$

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}(xf') = x\left(\frac{\partial f'}{\partial x}\cdot\frac{\partial x}{\partial x} + \frac{\partial f'}{\partial y}\cdot\cancel{\frac{\partial y}{\partial x}}^{0}\right) + f'$$

$$= x\frac{\partial f'}{\partial x} + f'$$

$$\therefore \boxed{u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = \frac{\partial^2 u}{\partial y^2} + \theta \sin n}$$

$$\Rightarrow xf'\left(x\frac{\partial f'}{\partial x} + f'\right) + \left(-x\frac{\partial f}{\partial x} - f\right)xf'' = xf''' + \theta \sin n$$

$$\Rightarrow f''' + ff'' - f'^2 = x\left(f'\frac{\partial f'}{\partial x} - f''\frac{\partial f}{\partial x}\right)$$

$$\Rightarrow \frac{\partial^3 f}{\partial y^3} + f\frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial f}{\partial y}\right)^2 + \theta \sin n = x\left(\frac{\partial f}{\partial x}\frac{\partial f'}{\partial x} - \frac{\partial^2 f}{\partial y^2}\frac{\partial f}{\partial x}\right)$$

$$\theta = \theta(x, y)$$

$$\frac{\partial \theta}{\partial x} = \frac{\partial \theta}{\partial \eta}\frac{\partial \eta}{\partial x} + \frac{\partial \theta}{\partial y}\frac{\partial y}{\partial x} = \frac{\partial \theta}{\partial \eta}$$

$$\frac{\partial \theta}{\partial y} = \frac{\partial \theta}{\partial x}\frac{\partial x}{\partial y} + \frac{\partial \theta}{\partial y}\frac{\partial y}{\partial y} = \frac{\partial \theta}{\partial y} = \theta'$$

$$\frac{\partial^2 \theta}{\partial y^2} = \theta''$$

$$u\frac{\partial \theta}{\partial x} + v\frac{\partial \theta}{\partial y} = \frac{1}{Pr}\frac{\partial^2 \theta}{\partial y^2} + \gamma\theta$$

$$\Rightarrow x f' \cdot \frac{\partial \theta}{\partial \eta} + \left(-x\frac{\partial f}{\partial \eta} - f\right)\theta' = \frac{1}{Pr}\theta'' + \gamma\theta$$

$$\Rightarrow \frac{1}{Pr}\theta'' + f\theta' + \gamma\theta = x\left(f'\frac{\partial \theta}{\partial \eta} - \theta'\frac{\partial f}{\partial \eta}\right)$$

$$\Rightarrow \frac{1}{Pr}\frac{\partial^2 \theta}{\partial y^2} + f\frac{\partial \theta}{\partial y} + \gamma\theta = x\left(\frac{\partial f}{\partial y}\frac{\partial \theta}{\partial x} - \frac{\partial \theta}{\partial y}\frac{\partial f}{\partial x}\right)$$